# FoodBridge

An Intelligent Food Rescue Coordination and Resource Allocation Platform

Software Engineering Laboratory (PTPS) · Final Project Demo

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# Problem statement and scope

Surplus food is discarded while the organizations that require it operate short of supply.

- Donations are coordinated through telephone calls and messaging applications, which delays collection and results in lost donations.
- Most donation platforms publish a listing and allocate the food to the first organization that claims it.
- They do not verify that the food can reach the recipient before its safe consumption window closes.
- They neither measure cumulative distribution across organizations nor record the reasoning behind a selection.

**Scope**

Scope of this release: four roles, one city, and the complete path from a registered donation to a confirmed delivery. Payments, inventory forecasting and route optimisation sit outside the scope.

# Differentiating features

Six design decisions separate FoodBridge from existing platforms and manual coordination.

| Design decision | Typical platforms | FoodBridge |
|---|---|---|
| Recipient selection | Allocated to the first organization that claims the listing | All eligible organizations are ranked by the decision engine |
| Food safety | Expiry time displayed to the claimant | Organizations that cannot receive the food in time are excluded |
| Equitable distribution | Not measured | Quantities received in the previous 30 days influence the score |
| Transparency | No recorded justification | A written explanation is stored with every allocation |
| Volunteer coordination | Arranged privately between donor and recipient | Volunteers are matched by distance, workload and experience |
| Accountability | Basic donation history | Every action is recorded in the audit log |

# Functional scope

Four actors and sixteen use cases, all implemented in the running system.




- The workflow proceeds from donation registration to a confirmed delivery, with the allocation decision recorded at each stage.
- Each role workspace exposes only the functions that the actor is authorised to perform.
- The engine operates on the server; the browser presents decisions and collects responses.

# Allocation methodology

The engine runs when food is registered. Four constraints apply first; six weighted factors then rank the rest.

**Verified organization** → **Sufficient free storage** → **Category accepted** → **Safe arrival feasible**

| Factor | Weight | Basis of measurement |
|---|---|---|
| Expiry urgency | 25% | Time remaining before the food becomes unsafe, weighted by the recipient's collection speed |
| Demand match | 20% | Correspondence between the donation quantity and the registered demand |
| Storage fit | 15% | Remaining storage capacity after intake |
| Proximity | 15% | Distance from the pickup point |
| Food preference | 15% | Whether the organization accepts the food category |
| Fair distribution | 10% | Quantity received by the organization during the previous 30 days |

# Original contribution: the food-safety arrival rule

A constraint introduced in this project. Food must be able to arrive while it remains safe to serve.

- The engine estimates the pickup to handover duration for each candidate organization from the travel distance and a handling allowance.
- The estimate is compared with the remaining safe window, retaining sixty minutes for serving.
- Organizations that cannot arrive in time are excluded, and the reason is stored on the allocation.
- Distance was initially treated as a scoring factor. Because late arrival is a safety failure rather than a lower priority, it was implemented as a hard constraint.

| Organization | Distance | Outcome |
|---|---|---|
| Sneha Sadan Orphanage | 11.6 km | Eligible, 15 minutes to spare |
| Sanjeevini Community Kitchen | 11.8 km | Eligible, 15 minutes to spare |
| Three organizations | 18 to 25 km | Excluded, 65 to 85 minutes required against 60 available |

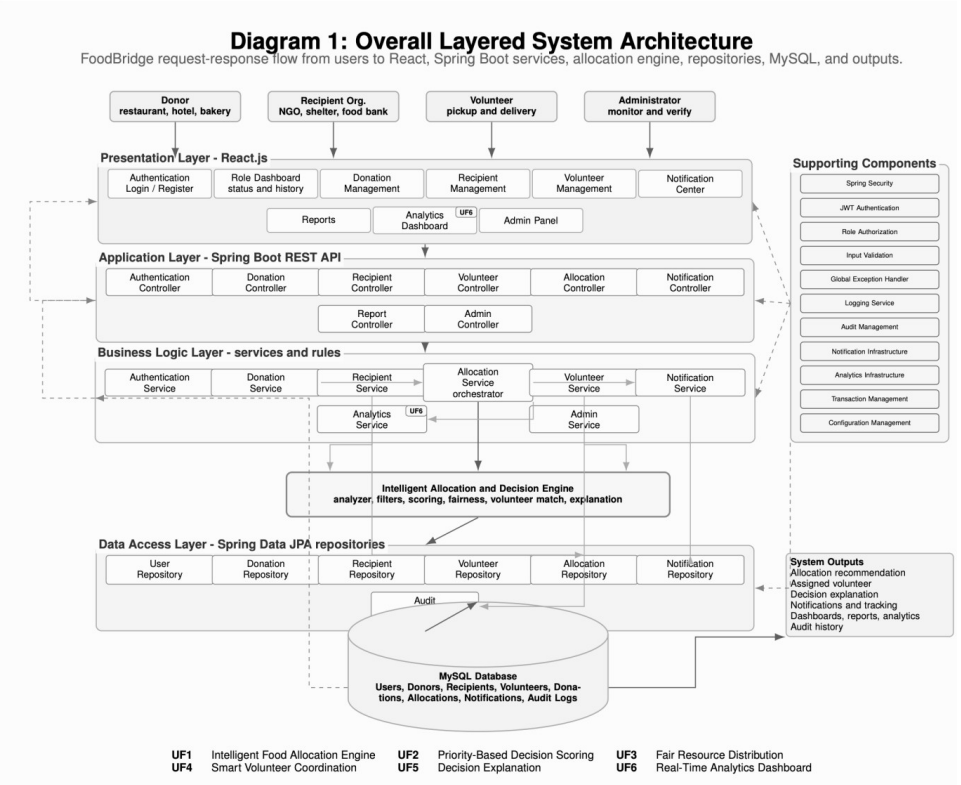
**Assumptions: 20 km/h urban travel, 10 minutes handling, 60 minutes serving buffer.**

# System architecture

The layered architecture submitted in the design phase, implemented as specified.

- Four layers: React presentation, Spring Boot REST application, business logic including the allocation engine, and JPA over MySQL.
- Authentication issues a JWT at login; the server verifies the role and the ownership of each record.
- The allocation engine is an independent service, which keeps decision rules separate from the domain classes.

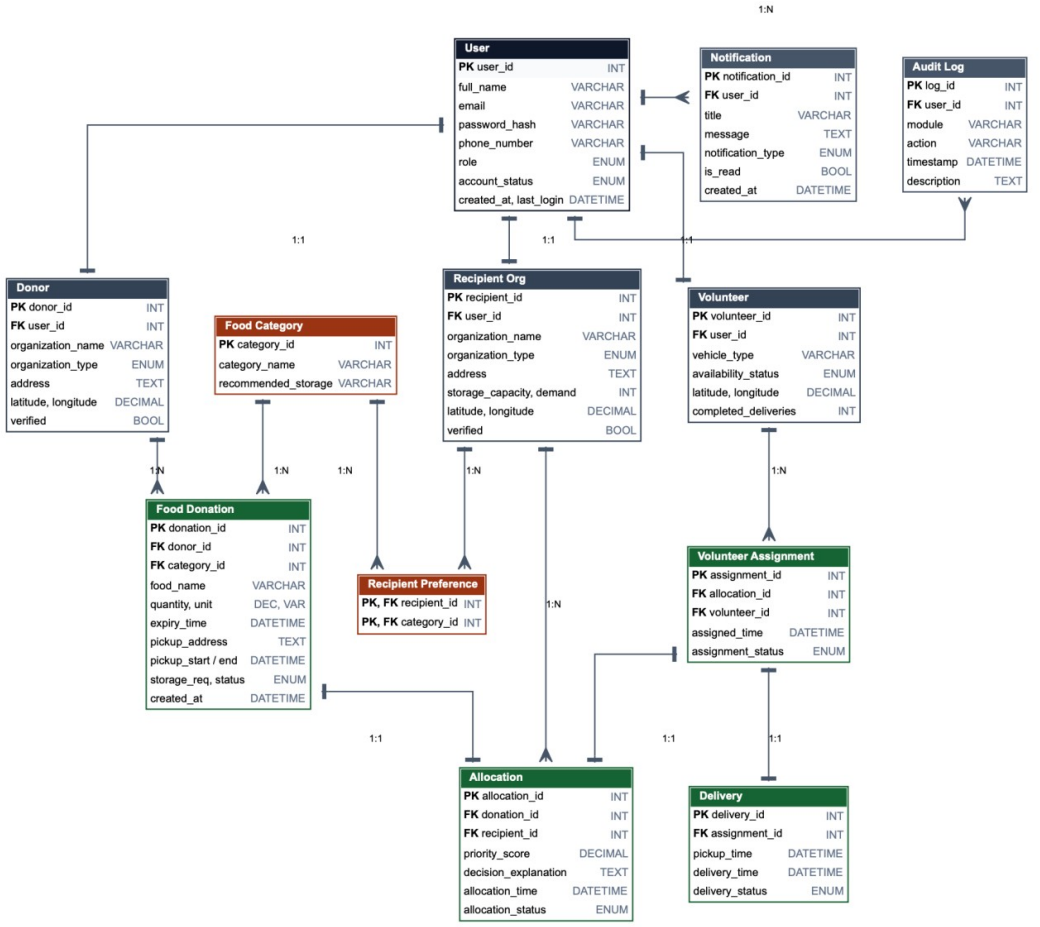
**System Architecture :**



# Database design

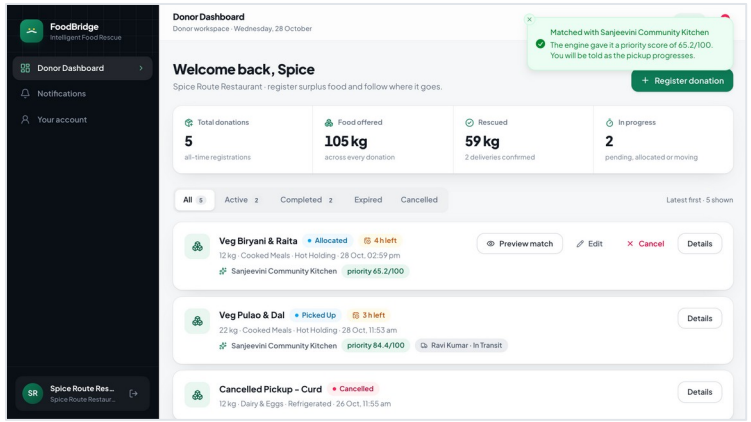
The entity relationship model submitted in the design phase, implemented as twelve tables.

- Twelve entities in third normal form.
- Allocation, assignment and delivery are modelled separately, so a decision, a task and a delivery are tracked independently.
- Recipient food preferences resolve the many to many relationship between organizations and categories.



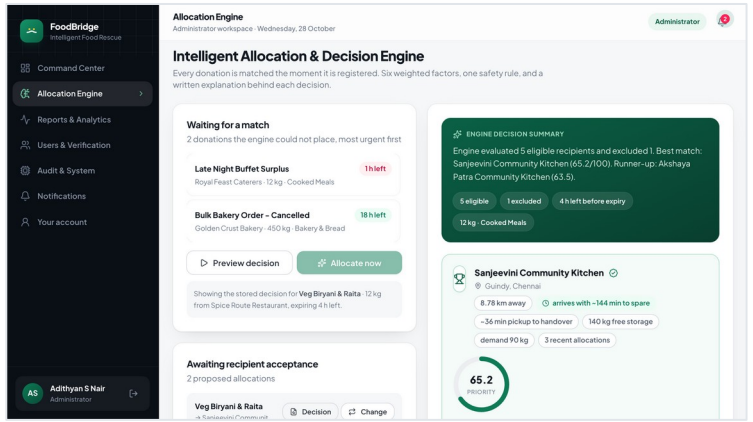
# Key features in the running application

A donation is matched on registration, the decision is explained, and the administrator can change the recipient.



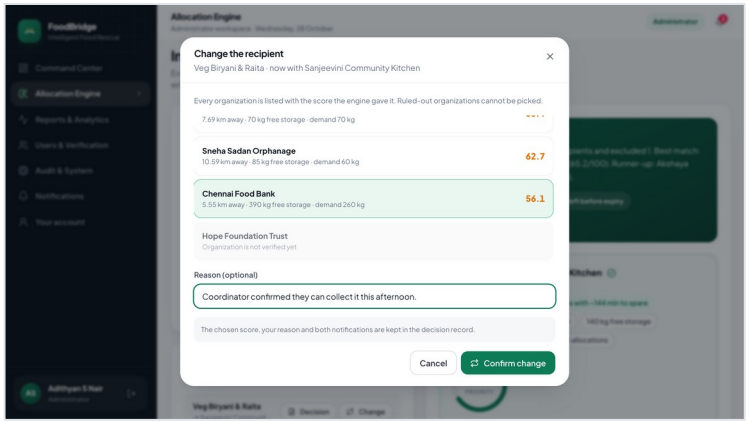

### Donor workspace

Food registered and matched immediately




### Allocation engine

Score, factors and safety check in one view




### Administrator override

Recipient changed, with a written reason

Demonstration path: the donor registers food, the administrator reviews or changes the decision, the recipient accepts, the volunteer delivers, and the recipient confirms receipt.

# Results and verification

Measurements taken from the running system with the seeded dataset.

**414 kg**
food rescued

**1,035**
meals equivalent

**84.6%**
allocations accepted

**55 min**
pickup to confirmation

**15 of 15**
automated tests pass

**6**
recipient organisations

- Fifteen automated tests cover the eligibility constraints, scoring, fairness, the arrival rule, access control, automatic matching, override and volunteer assignment.
- Eighteen API checks confirm the role restrictions, including rejection of unauthorised requests with 401 and 403 responses.
- A complete rescue has been executed end to end, including a recipient change, delivery and confirmation.

# Challenges, limitations and future improvements

What was difficult to get right, where the current release stops, and what comes next.

### Challenges faced

- Food safety had to become a hard filter. A good score cannot excuse food that arrives after its safe window closes.
- Six factors had to stay explainable. Every weight, exclusion and override is visible in the decision record.
- Four roles share one workflow, so the API checks access and record ownership for every request.
- A rejection or recipient change has to leave the donation, volunteer and storage figures consistent.

### Limitations and next steps

- Distances use straight lines and one city speed. Next: road routing, traffic and per-vehicle speeds.
- Locations are coordinates entered on sign-up. Next: a map picker and live volunteer position.
- The system runs locally on MySQL. Next: a hosted deployment with backups and monitoring.
- Notifications are in-app. Next: SMS or WhatsApp for organizations that do not keep the site open.